Siddhartha Prasad

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I am a programming languages researcher interested in the human aspects of programming and formal reasoning. I study not only how PL and FM techniques can solve problems programmers face, but also how programmers' experiences can inform the design of our languages, logics, and software tools. My research is informed by my time as an engineer: I have written code that doesn't do what I want it to, and I want to spare everyone else the indignity.

Education

Ph.D. in Computer Science, **Brown University**, Providence, RI *Jan 2022 – May 2027 (expected)*

Advisor: Shriram Krishnamurthi

B.S. in Computer Science and Mathematics, **Tufts University**, Medford, MA *May 2016*

Advisor: Kathleen Fisher · *Summa cum Laude; High Thesis Honors*

Employment

Apple · Research Intern · Human-Centered Machine Learning · Seattle, WA *May – Sep 2023*

- Designed a formal-methods-inspired technique that prompts an LLM to evaluate candidate outputs against task-specific quality criteria and iteratively refine them without fine-tuning.
- Applied the technique in ContextQ, a system that generates questions to support parent-child dialogue during shared reading.

Microsoft Redmond, WA

Software Engineer II · Applied AI *Apr 2018 – Sep 2021*

- Built the container platform for Azure Cognitive Services, enabling customers with sensitive data to run AI workloads on-premises, at the edge, or in fully disconnected environments.
- Designed a deployment framework that allowed the same Docker containers to run across Azure-hosted and customer-managed environments, abstracting differences in CPU architecture, connectivity, API-key validation, logging, and storage.
- Operated production infrastructure for the cloud service, including on-call responsibility.

Software Engineer · Developer Ecosystem Platform *Aug 2016 – Mar 2018*

- Designed and shipped XAML input APIs across Windows 10, Windows Phone, and Xbox, giving developers a consistent model for keyboard, gamepad, and ink interaction.
- Built accessibility and navigation features, including XAML access keys, so applications could expose commands and behave consistently across keyboard and gamepad input.

Software Engineering Intern · Developer Ecosystem Platform *May – Aug 2015*

INRIA · Research Intern · Parsifal · Saclay, France *Jun – Jul 2014*

- Researched correctness certificates for first-order term-rewriting systems.

Publications

Diagramming Program Values by Spatial Refinement, **PLDI 2026**
Siddhartha Prasad, Michael Tu, Karan Kashyap, Tim Nelson, Shriram Krishnamurthi

Meaningful Human-in-the-Loop Checking of GenAI Synthesis for Restricted Languages, **ECOOP 2026**
Siddhartha Prasad, Skyler Austen, Kathi Fisler, Shriram Krishnamurthi

hS: Speculative Script Reordering at Subprocess Granularity, **OSDI 2026**
Georgios Liargkovas, Di Jin, Tianyu (Ezri) Zhu, Dan Liu, A. Bolun Thompson, Anirudh Narsipur, Seong-Heon Jung, *Siddhartha Prasad*, Diomidis Spinellis, Michael Greenberg, Konstantinos Kallas, Nikos Vasilakis

A Misconception-Driven Adaptive Tutor for Linear Temporal Logic, **CAV 2025**
[Distinguished Paper]
Siddhartha Prasad, Ben Greenman, Tim Nelson, Shriram Krishnamurthi

Lightweight Diagramming for Lightweight Formal Methods: A Grounded Language Design, **ECOOP 2025**
[Distinguished Paper; Distinguished Artifact]
Siddhartha Prasad, Ben Greenman, Tim Nelson, Shriram Krishnamurthi

Misconceptions in Finite-Trace and Infinite-Trace Linear Temporal Logic, **FM 2024**
Ben Greenman, *Siddhartha Prasad*, Antonio Di Stasio, Shufang Zhu, Giuseppe De Giacomo, Shriram Krishnamurthi, Marco Montali, Tim Nelson, Milda Zizyte

ContextQ: Generated Questions to Support Meaningful Parent-Child Dialogue While Co-Reading, **IDC 2024**
Griffin Dietz Smith, *Siddhartha Prasad*, Matt J Davidson, Leah Findlater, R Benjamin Shapiro

Forge: A Tool and Language for Teaching Formal Methods, **OOPSLA 2024**
Tim Nelson, Ben Greenman, *Siddhartha Prasad*, Tristan Dyer, Ethan Bove, Qianfan Chen, Charles Cutting, Thomas Del Vecchio, Sidney LeVine, Julianne Rudner, Ben Ryjikov, Alexander Varga, Andrew Wagner, Luke West, Shriram Krishnamurthi

Conceptual Mutation Testing for Student Programming Misconceptions, ‹**Programming**› **2024**
Siddhartha Prasad, Ben Greenman, Tim Nelson, Shriram Krishnamurthi

Generating Programs Trivially: Student Use of Large Language Models, **CompEd 2023**
Siddhartha Prasad, Ben Greenman, Tim Nelson, Shriram Krishnamurthi

Making Hay from Wheats: A Classsourcing Method to Identify Misconceptions, **Koli Calling 2022**
Siddhartha Prasad, Ben Greenman, Tim Nelson, John Wrenn, Shriram Krishnamurthi

Large-Scale Intelligent Microservices, **IEEE BigData 2020**
Mark Hamilton, Nick Gonsalves, Christina Lee, Anand Raman, Brendan Walsh, *Siddhartha Prasad*, Dalitso Banda, Lucy Zhang, Lei Zhang, William T Freeman

Invited Talks

Apr 2026	Eventually, Understanding: Operationalizing Misconceptions in LTL Brandeis University
Mar 2026	Eventually, Understanding: Operationalizing Misconceptions in LTL IARCS Verification Seminar Series
Jan 2026	Lightweight Diagramming by Spatial Specification MIT HCI Seminar

Jan 2024 Understanding Problems Before Programming
IIT Gandhinagar
Mar 2023 Understanding Problems Before Programming
Ashoka University

Teaching

Teaching Assistant, **Brown University**

Logic for Systems 2022–2026
Programming Languages 2025

Teaching Assistant, **Tufts University**

Programming Languages 2015–2016
Algorithms 2014

Mentorship

Summer 2026 **Owen Duckham**, Diagramming for Lean
Brown University
Summer 2025 **Michael Tu**, Spatial refinement for diagramming program values
Brown University
Winter 2019 **Patrick Kao**, Windows emulation on Linux
Microsoft

Service & Volunteering

2026 Artifact Reviewer, The Art, Science, and Engineering of Programming
2017 Volunteer Alumni Interviewer, Tufts Admissions Network

Affiliations & Certifications

Sheridan Teaching Seminar Certificate · Brown University
Tau Beta Pi, The Engineering Honor Society · Tufts University